Welcome to Amazon SimpleDB

This is the Developer Guide for *Amazon SimpleDB*. This guide provides a conceptual overview of Amazon SimpleDB, programming reference material, and a detailed API reference.

Amazon SimpleDB is a web service for running queries on structured data in real time. This service works in close conjunction with Amazon Simple Storage Service (Amazon S3) and Amazon Elastic Compute Cloud (Amazon EC2), collectively providing the ability to store, process and query data sets in the cloud. These services are designed to make web-scale computing easier and more cost-effective for developers.

How Do I...?	Relevant Sections
Learn if Amazon SimpleDB is right for my use case	Amazon SimpleDB Detail Page
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Learn how to tune Amazon SimpleDB queries	Tuning Queries
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Introduction to Amazon SimpleDB

Topics

- Features
- How Amazon Charges for Amazon SimpleDB
- Viewing Your Bill

This introduction to Amazon SimpleDB is intended to give you a detailed summary of this web service. After reading this section, you should have a good idea of what it offers and how it can fit in with your business.

Traditionally, the type of functionality provided by Amazon SimpleDB has been accomplished with a clustered relational database that requires a sizable upfront investment, brings more complexity than is typically needed, and often requires a DBA to maintain and administer. In contrast, Amazon SimpleDB is easy to use and provides the core functionality of a database - real-time lookup and simple querying of structured data - without the operational complexity. Amazon SimpleDB requires no schema, automatically indexes your data and provides a simple API for storage and access. This eliminates the administrative burden of data modeling, index maintenance, and performance tuning. Developers gain access to this functionality within Amazon's proven computing environment, are able to scale instantly, and pay only for what they use.

Features

Following are some of the major Amazon SimpleDB attributes:

Features

- **Simple to use—**Amazon SimpleDB provides streamlined access to the lookup and query functions that traditionally are achieved using a relational database cluster while leaving out other complex, often-unused database operations.

  The service allows you to quickly add data and easily retrieve or edit that data through a simple set of API calls. Accessing these capabilities through a web service also eliminates the complexity of maintaining and scaling these operations

- **Flexible—**With Amazon SimpleDB, it is not necessary to pre-define all of the data formats you will need to store; simply add new attributes to your Amazon SimpleDB data set when needed, and the system will automatically index your data accordingly.

  The ability to store structured data without first defining a schema provides developers with greater flexibility when building applications.
- **Scalable—**Amazon SimpleDB allows you to easily scale your application. You can quickly create new domains as your data grows or your request throughput increases.

  Currently, you can store up to 10 GB per domain and you can create up to 250 domains.
- **Fast—**Amazon SimpleDB provides quick, efficient storage and retrieval of your data to support high performance web applications.
- **Reliable—**The service runs within Amazon's high-availability data centers to provide strong and consistent performance.

  To prevent data from being lost or becoming unavailable, your fully indexed data is stored redundantly across multiple servers and data centers.
- **Designed for use with other Amazon Web Services—**Amazon SimpleDB is designed to integrate easily with other web-scale services such as Amazon EC2 and Amazon S3.

  For example, developers can run their applications in Amazon EC2 and store their data objects in Amazon S3. Amazon SimpleDB can then be used to query the object metadata from within the application in Amazon EC2 and return pointers to the objects stored in Amazon S3.
- **Inexpensive—**Amazon SimpleDB passes on to you the financial benefits of Amazon's scale. You pay only for resources you actually consume.

  Compare this with the significant up-front expenditures traditionally required to obtain software licenses and purchase and maintain hardware, either in-house or hosted. This frees you from many of the complexities of capacity planning, transforms large capital expenditures into much smaller operating costs, and eliminates the need to over-buy "safety net" capacity to handle periodic traffic spikes.

How Amazon Charges for Amazon SimpleDB

Amazon SimpleDB pricing is based on your actual usage. Your usage is measured and rounded up to the nearest cent.

Amazon SimpleDB charges you for the following types of usage:

- **Structured Data Storage—**Measures the size of your billable data by adding the raw byte size of the data you upload + 45 bytes of overhead for each item, attribute name and attribute-value pair.
- **Data Transfer—**Measures the amount of data transferred for every operation.

  Data transferred between Amazon SimpleDB and other Amazon Web Services (e.g., Amazon S3, Amazon EC2, Amazon SQS, and others) is free of charge.
- **Machine Utilization—**Measures the *machine utilization* of each request and charges based on the amount of machine capacity used to complete the particular request (SELECT, GET, PUT, etc.).

> **Note**
>
> Amazon Web Services provides a free tier of Amazon SimpleDB usage. The free tier is a monthly offer. Free usage does not accumulate.
> Any data stored as part of the free tier program must be actively used. If a domain is not accessed for a period of 6 months, it will be subject to removal at the discretion of Amazon Web Services.

Storage

You are charged for the amount of storage your data uses each month which can be considered an average of the month. For example, if you use one gigabyte for the month, you are charged for one gigabyte of storage. If you use zero gigabytes for the first half of the month and two gigabytes for the second half, you are also charged for one gigabyte of storage.

Several times each day, Amazon SimpleDB measures the amount of storage used by all of the objects in your account. Amazon SimpleDB stores this information in byte-hours and averages it with all other recorded measurements at the end of the billing cycle.

Data Transfer

Amazon charges you for the amount of data transferred into and out of Amazon SimpleDB. For every operation, Amazon SimpleDB monitors the amount of data sent and received and records the data. Once per hour, the usage total is recorded to your account. This information is stored and totaled at the end of the billing cycle.

Machine Utilization

For each successful request, Amazon SimpleDB charges you for the amount of machine capacity used to complete that request.

You are also charged for any request that fails with an HTTP 4xx error. For example, if Amazon SimpleDB cannot authorize a request, you will receive an HTTP 403 error (Forbidden) and incur a machine utilization charge for the failed request.

Viewing Your Bill

You can view the charges for your current billing period at any time by going to the AWS Portal.

To view your activity

1. Log in to your AWS account.
2. Move the pointer over **Your Web Services Account**.
3. Click **Account Activity**.

   A list of services to which you subscribe appears.
4. Locate the Amazon SimpleDB service.

Billing Component	Description
View/Edit Service Button	Enables you to view or change settings associated with the Amazon SimpleDB service.
Machine Hour Usage	Shows the machine hour usage cost, current number of hours consumed, and billing for the current cycle.
Data Transferred	Shows the data transfer cost, current amount of data transferred, and billing for the current cycle.
Storage	Shows the storage usage cost, average amount of storage consumed, and billing for the current cycle.
Usage Report	Shows detailed data used to calculate your bill.

Amazon SimpleDB Concepts

Topics

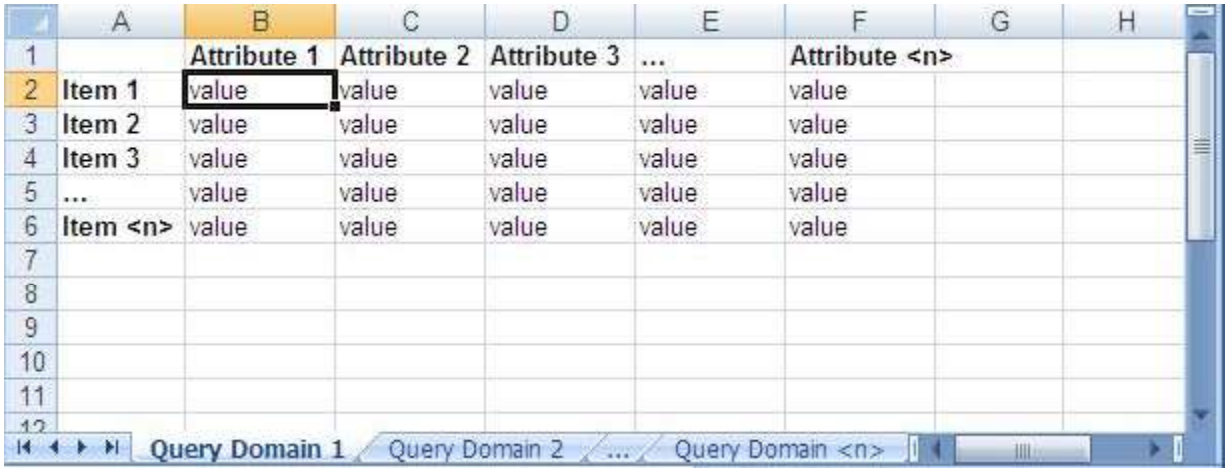
- Data Model
- Operations
- API Summary
- Consistency
- Limits
- Data Set Partitioning
- AWS Identity and Access Management

This section describes the key concepts that you should understand before using Amazon SimpleDB.

Data Model

When using Amazon SimpleDB, you organize your structured data in domains within which you can put data, get data, or run queries.

Domains consist of items which are described by *attribute* name-value pairs. Consider the spreadsheet model shown in the following image.



The components correspond to each part of a spreadsheet:

- **Customer Account—**Represented by the entire spreadsheet, it refers to the Amazon Web Services *account* to which all domains are assigned.
- **Domains—**Represented by the domain worksheet tabs at the bottom of the spreadsheet, domains are similar to tables that contain similar data.

  You can execute queries against a domain, but cannot execute queries across different domains.
- **Items—**Represented by rows, items represent individual objects that contain one or more attribute name-value pairs.
- **Attributes—**Represented by columns, attributes represent categories of data that can be assigned to items.
- **Values—**Represented by cells, values represent instances of attributes for items. An attribute can have multiple values.

Unlike a spreadsheet, however, multiple values can be associated with a cell. For example, an item can have both the color value *red* and *blue*. Additionally, Amazon SimpleDB does not require the presence of specific attributes. You can create a single domain that contains completely different product types. For example, the following table contains clothing, automotive parts, and motorcycle parts.

ID	Category	Subcat.	Name	Color	Size	Make	Model
Item_01	Clothes	Sweater	Cathair Sweater	Siamese	Small, Medium, Large		
Item_02	Clothes	Pants	Designer Jeans	Paisley Acid Wash	30x32, 32x32, 32x34		
Item_03	Clothes	Pants	Sweatpants	Blue, Yellow, Pink	Large		
Item_04	Car Parts	Engine	Turbos			Audi	S4
Item_05	Car Parts	Emissions	02 Sensor			Audi	S4

ID	Category	Subcat.	Name	Color	Size	Make	Model
Item_06	Motorcycle Parts	Bodywork	Fender Eliminator	Blue		Yamaha	R1
Item_07	Motorcycle Parts, Clothing	Clothing	Leather Pants	Black	Small, Medium, Large		

Regardless of how you store your data, Amazon SimpleDB automatically indexes your data for quick and accurate retrieval.

Operations

The following describes components of Amazon SimpleDB operations.

- **Subscriber—**Any application, script, or software making a call to the Amazon SimpleDB service.

  The AWS Access Key ID uniquely identifies each subscriber for billing and metering purposes.
- **Amazon SimpleDB Request—**A single web service API call and its associated data that the subscriber sends to the Amazon SimpleDB service to perform one or more operations.
- **Amazon SimpleDB Response—**The response and any results returned from the Amazon SimpleDB service to the subscriber after processing the request.

  The AWS Platform handles authentication success and failure; failed requests are not sent to the Amazon SimpleDB service.

API Summary

The Amazon SimpleDB service consists of a small group of API calls that provide the core functionality you need to build your application. See Operations in the API Reference chapter for detailed descriptions of each option.

- **CreateDomain—**Create domains to contain your data; you can create up to 250 domains. If you require additional domains, go to https://console.aws.amazon.com/support/home#/case/create?issueType=service-limit-increase&limitType=service-code-simpledb-domains.

- **DeleteDomain—**Delete any of your domains
- **ListDomains—**List all domains within your account
- **PutAttributes—**Add, modify, or remove data within your Amazon SimpleDB domains
- **BatchPutAttributes—**Generate multiple put operations in a single call
- **DeleteAttributes—**Remove items, attributes, or attribute values from your domain
- **BatchDeleteAttributes—**Generate multiple delete operations in a single call
- **GetAttributes—**Retrieve the attributes and values of any item ID that you specify
- **Select—**Query the specified domain using a SQL SELECT expression
- **DomainMetadata—**View information about the domain, such as the creation date, number of items and attributes, and the size of attribute names and values

Consistency

Amazon SimpleDB keeps multiple copies of each domain. A successful write (using PutAttributes, BatchPutAttributes, DeleteAttributes, BatchDeleteAttributes, CreateDomain, or DeleteDomain) guarantees that all copies of the domain will durably persist.

Amazon SimpleDB supports two read consistency options: eventually consistent read and consistent read.

An eventually consistent read (using Select or GetAttributes) might not reflect the results of a recently completed write (using PutAttributes, BatchPutAttributes, DeleteAttributes, or BatchDeleteAttributes). Consistency across all copies of the data is usually reached within a second; repeating a read after a short time should return the updated data.

A consistent read (using Select or GetAttributes with `ConsistentRead=true`) returns a result that reflects all writes that received a successful response prior to the read.

By default, GetAttributes and Select perform an eventually consistent read.

The following table describes the characteristics of eventually consistent read and consistent read.

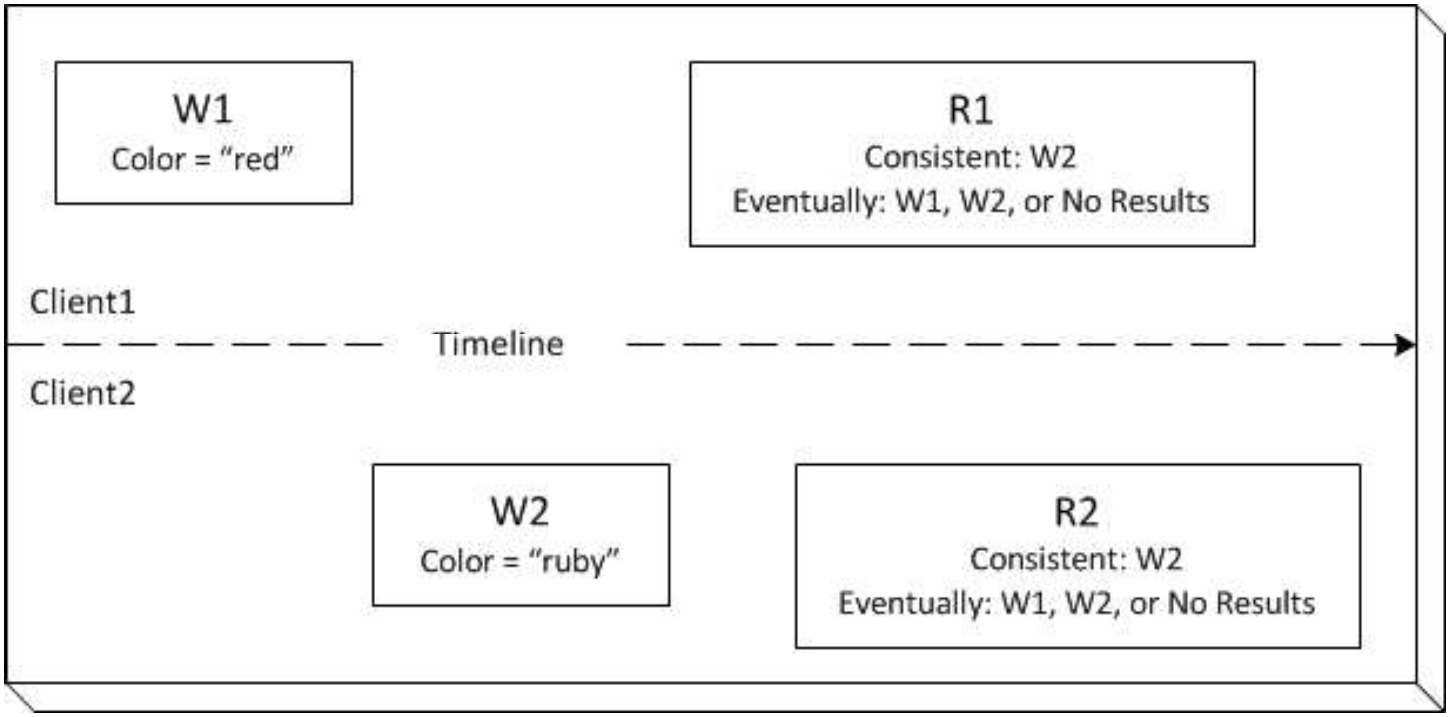
Eventually Consistent Read	Consistent Read
Stale reads possible	No stale reads
Lowest read latency	Potential higher read latency

Eventually Consistent Read	Consistent Read
Highest read throughput	Potential lower read throughput

Concurrent Applications

This section provides examples of eventually consistent and consistent read requests when multiple clients are writing to the same items. Whenever you have multiple clients writing to the same items, implement some concurrently control mechanism, such as timestamp ordering, to ensure you are getting the data you want.

In this example, both W1 (write 1) and W2 (write 2) complete (receive a successful response from the server) before the start of R1 (read 1) and R2 (read 2). For a consistent read, R1 and R2 both return `color = ruby`. For an eventually consistent read, R1 and R2 might return `color = red`, `color = ruby`, or no results, depending on the amount of time that has elapsed.

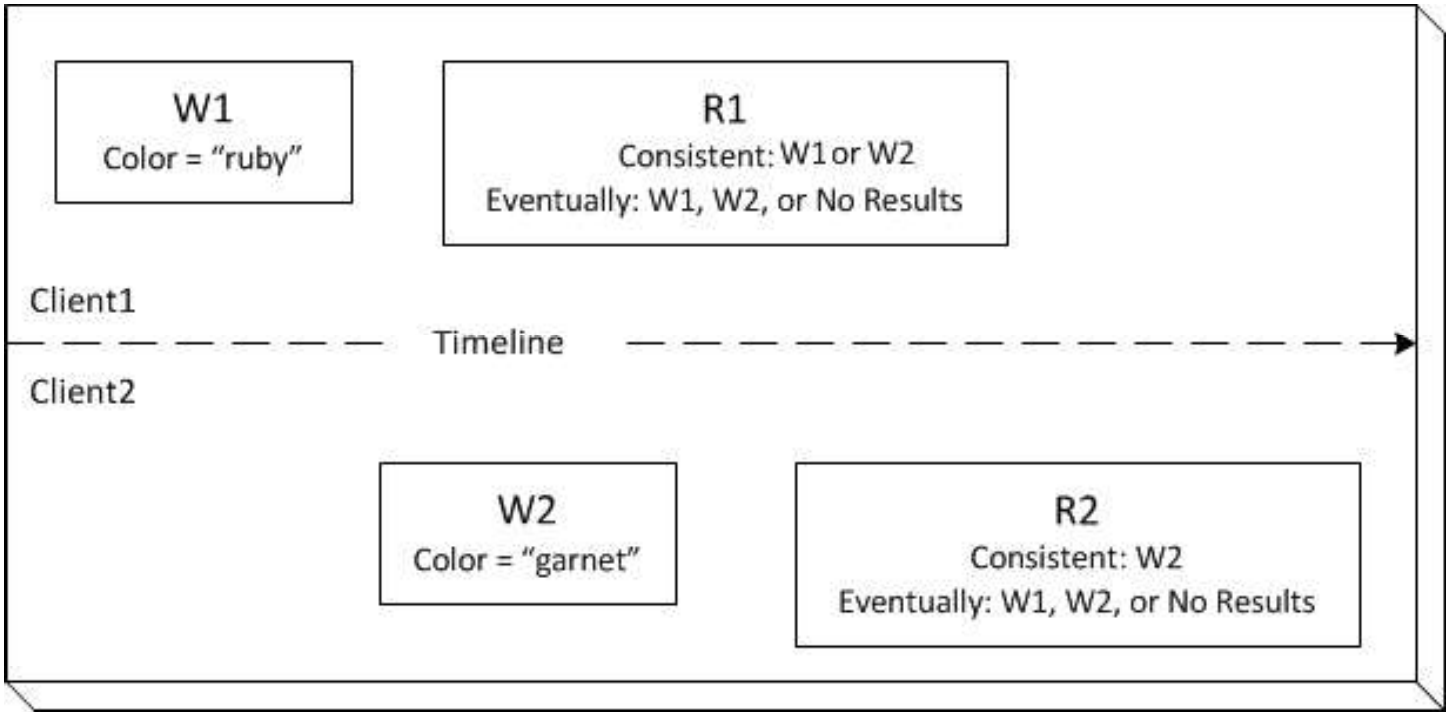


In the next example, W2 does not complete before the start of R1. Therefore, R1 might return `color = ruby` or `color = garnet` for either a consistent read or an eventually consistent read.

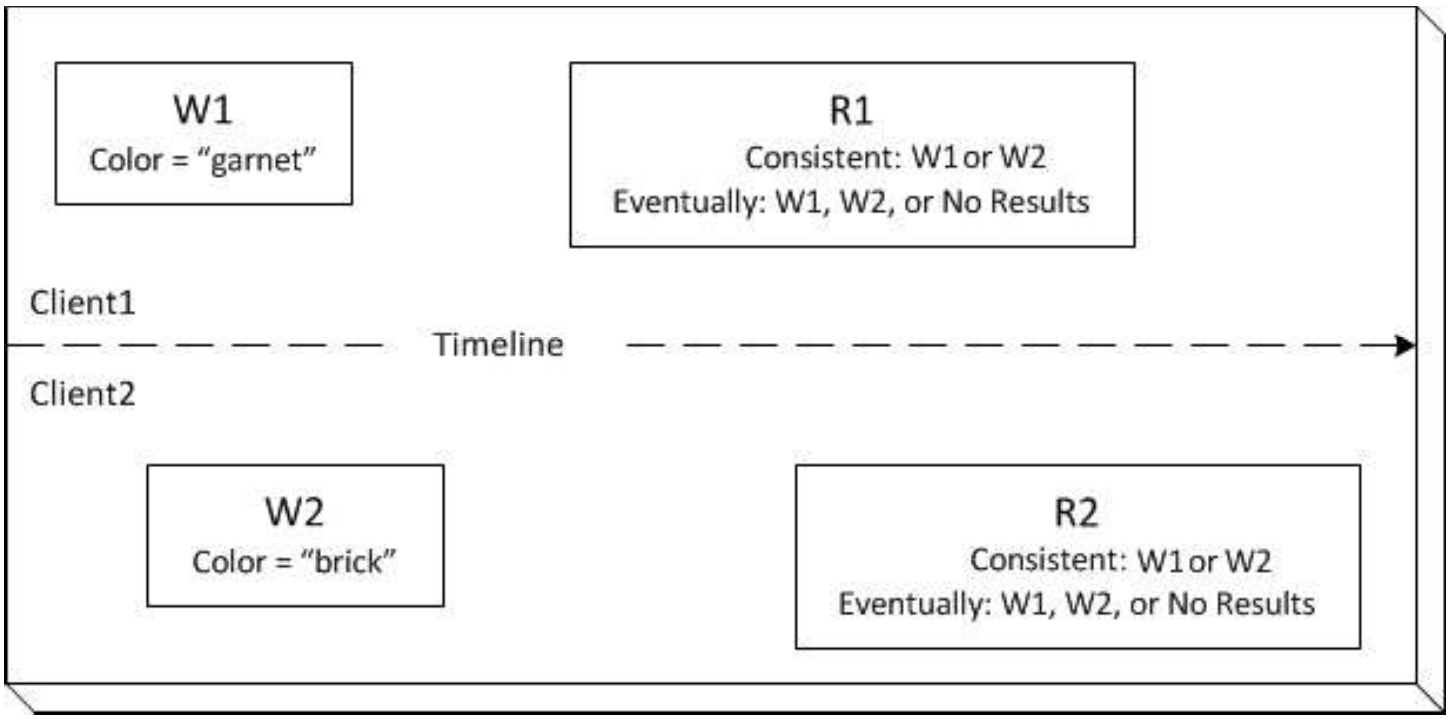
Data is distributed among several servers. If R1 is sent to one server that does not have the W2 values, yet, then R1 returns W1 values. Also, depending on the amount of time that has elapsed, an eventually consistent read might return no results.

> **Note**
>
> If a failure occurs during the second write operation (W2), the value might change depending on when in the operation the failure occurs.

For a consistent read, R2 returns `color = garnet`. For an eventually consistent read, R2 might return `color = ruby`, `color = garnet`, or no results depending on the amount of time that has elapsed.



In the last example, Client 2 submits W2 before Amazon SimpleDB completes W1, so the outcome of the final value is unknown (`color = garnet` or `color = brick`). Any subsequent reads (consistent read or eventually consistent) might return either value. Also, depending on the amount of time that has elapsed, an eventually consistent read might return no results.



Limits

Following is a table that describes current limits within Amazon SimpleDB.

Parameter	Restriction
Domain size	10 GB per domain
Domain size	1 billion attributes per domain
Domain name	3-255 characters (a-z, A-Z, 0-9, '_', '-', and '.')
Domains per account	250
Attribute name-value pairs per item	256
Attribute name length	1024 bytes
Attribute value length	1024 bytes
Item name length	1024 bytes

Parameter	Restriction
Attribute name, attribute value, and item name allowed characters	All UTF-8 characters that are valid in XML documents. Control characters and any sequences that are not valid in XML are returned Base64-encoded. For more information, see Working with XML-Restricted Characters.
Attributes per `PutAttributes` operation	256
Attributes requested per `Select` operation	256
Items per `BatchDeleteAttributes` operation	25
Items per `BatchPutAttributes` operation	25
Maximum items in `Select` response	2500
Maximum query execution time	5 seconds
Maximum number of unique attributes per `Select` expression	20
Maximum number of comparisons per `Select` expression	20
Maximum response size for `Select`	1MB

Data Set Partitioning

Amazon SimpleDB is designed to support highly parallel applications. To improve performance, you can partition your dataset among multiple domains to parallelize queries and have them operate on smaller individual datasets. Although you can only execute a single query against